

Activity Options

Course Routines + Claim–Evidence–Reasoning

Two reusable full sets

24 questions each

LEARNING TARGET

I can successfully participate in physics class by following course procedures and using evidence-based reasoning to explain physical phenomena.

I CAN STATEMENTS

- I can follow the procedures and routines used in physics class.
- I can identify the parts of a scientific explanation: Claim, Evidence, and Reasoning (CER).
- I can make a clear claim that answers a question about a physical phenomenon.
- I can select and use relevant observations or data as evidence to support my claim.
- I can explain how my evidence supports my claim using scientific reasoning.

PROJECTION / WHITEBOARD OPTIONS

- [Whiteboard Indy](#) — [Set 1](#) · [Set 2](#)
- [Whiteboard Partners](#) — [Set 1](#) · [Set 2](#)
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PRINTABLE HANDOUTS

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TEACHER GUIDE

[Open teacher guide](#)

Whiteboard Indy

Course Routines + CER

[← Activity Options](#)

LEARNING TARGET

I can successfully participate in physics class by following course procedures and using evidence-based reasoning to explain physical phenomena.

STRUCTURE

Individual → Independent practice

SETUP

Each student gets a whiteboard and one marker.

DIRECTIONS

1. Try the projected physics problem before asking for an answer.
2. Write large enough that your work or CER labels can be read across the room.
3. Use Notes and the physical situation before searching for a formula.
4. Hold up your board when asked and be ready to explain the evidence or relationship you used.

GOAL

Make the physics thinking visible: identify the phenomenon, use relevant evidence, and explain the physical relationship.

QUESTION SETS

[Set 1](#) · [Set 2](#)

Looks Like Success

- Everyone has visible thinking.
- Students refer to the phenomenon/data, not just an answer.
- Mistakes are revised with an explanation.

Doesn't Look Like

- Blank boards while waiting for someone else.
- Copying a neighbor without discussing evidence.
- Plugging numbers into an equation with no physical story.

Whiteboard Partners

Course Routines + CER

[← Activity Options](#)

LEARNING TARGET

I can successfully participate in physics class by following course procedures and using evidence-based reasoning to explain physical phenomena.

STRUCTURE

Partners → Solve, compare, explain

SETUP

Partners each need a whiteboard and marker.

DIRECTIONS

1. Both partners attempt the problem independently.
2. Compare claims, evidence choices, calculations, or reasoning.
3. If work differs, explain one step or physical assumption at a time.
4. Revise your own board when needed.
5. Be ready for either partner to explain the pair's reasoning.

GOAL

Make the physics thinking visible: identify the phenomenon, use relevant evidence, and explain the physical relationship.

QUESTION SETS

[Set 1](#) · [Set 2](#)

Looks Like Success

- Both partners have their own work.
- Partners compare reasoning, not just final answers.
- Either partner can explain the physical situation.

Doesn't Look Like

- One student solves while the other watches.
- Partners erase differences without discussing them.

- Only one person can explain the answer.

Rally Coach

Course Routines + CER

[← Activity Options](#)

LEARNING TARGET

I can successfully participate in physics class by following course procedures and using evidence-based reasoning to explain physical phenomena.

STRUCTURE

Partners → Solver + Coach

SETUP

Partners share a projected/question set. One begins as Solver and the other as Coach.

DIRECTIONS

1. Solver works and explains the physical story aloud.
2. Coach asks questions such as “What is the phenomenon?” “What is your evidence?” or “What does that equation mean here?”
3. Coach does not take over the problem.
4. Switch roles after each problem.

GOAL

Make the physics thinking visible: identify the phenomenon, use relevant evidence, and explain the physical relationship.

QUESTION SETS

[Set 1](#) · [Set 2](#)

Looks Like Success

- Solver explains, not just writes.
- Coach asks questions rather than supplying steps.
- Roles switch consistently.

Doesn't Look Like

- Coach grabs the board and solves.
- Solver recites numbers without meaning.

- One person keeps the same role.

Speed Dating Physics

Course Routines + CER

[← Activity Options](#)

LEARNING TARGET

I can successfully participate in physics class by following course procedures and using evidence-based reasoning to explain physical phenomena.

STRUCTURE

Timed partner rounds → Explain, compare, rotate

SETUP

Students face a partner with a whiteboard or paper. Teacher controls rotation.

DIRECTIONS

1. Attempt the displayed problem.
2. Compare work with your partner.
3. Each person explains one useful observation, relationship, or check.
4. Rotate only when signaled.
5. Begin the next problem with the new partner.

GOAL

Make the physics thinking visible: identify the phenomenon, use relevant evidence, and explain the physical relationship.

QUESTION SETS

[Set 1](#) · [Set 2](#)

Looks Like Success

- Both students attempt each problem.
- Conversation reaches evidence/reasoning.
- Rotations happen quickly on signal.

Doesn't Look Like

- Students wait for a partner to do the thinking.
- Conversation stops at "same answer."

- Students move before the signal.

Showdown

Course Routines + CER

[← Activity Options](#)

LEARNING TARGET

I can successfully participate in physics class by following course procedures and using evidence-based reasoning to explain physical phenomena.

STRUCTURE

Individual think → Simultaneous reveal → Team discussion

SETUP

Groups of 3–4. Each student needs a whiteboard or paper.

DIRECTIONS

1. Everyone answers independently and keeps the response hidden.
2. Reveal on the teacher's signal.
3. If answers differ, compare the physical assumptions/evidence.
4. Each student corrects their own work.
5. Prepare one team explanation.

GOAL

Make the physics thinking visible: identify the phenomenon, use relevant evidence, and explain the physical relationship.

QUESTION SETS

[Set 1](#) · [Set 2](#)

Looks Like Success

- Everyone thinks before the reveal.
- Differences lead to discussion of physics.
- Students correct their own work.

Doesn't Look Like

- Answers are shown early.
- The group votes without checking evidence.

- One student fixes everyone else's work.

Think, Trade, Agree

Course Routines + CER

[← Activity Options](#)

LEARNING TARGET

I can successfully participate in physics class by following course procedures and using evidence-based reasoning to explain physical phenomena.

STRUCTURE

Think alone → Trade reasoning → Reach agreement

SETUP

Partners each need their own work surface.

DIRECTIONS

1. Think and answer independently.
2. Trade explanations with your partner.
3. Ask at least one clarifying question.
4. Agree on the reasoning/evidence, not just the answer.
5. Revise your own response if your thinking changes.

GOAL

Make the physics thinking visible: identify the phenomenon, use relevant evidence, and explain the physical relationship.

QUESTION SETS

[Set 1](#) · [Set 2](#)

Looks Like Success

- Each student brings an initial idea.
- Partners ask clarifying questions.
- Agreement includes a reason.

Doesn't Look Like

- One student copies the other.
- Partners only compare final words/numbers.

- No one revises after discussion.

Round Table

Course Routines + CER

[← Activity Options](#)

LEARNING TARGET

I can successfully participate in physics class by following course procedures and using evidence-based reasoning to explain physical phenomena.

STRUCTURE

Small group → One contribution at a time

SETUP

Groups of 3–4 share one large whiteboard or paper while each student keeps personal notes.

DIRECTIONS

1. One student records the group claim/answer.
2. Next student adds evidence or a physical relationship.
3. Continue around the group; each person must add, question, or revise something.
4. The group finishes by checking whether the explanation answers the original phenomenon question.

GOAL

Make the physics thinking visible: identify the phenomenon, use relevant evidence, and explain the physical relationship.

QUESTION SETS

[Set 1](#) · [Set 2](#)

Looks Like Success

- Every person contributes to the explanation.
- Students build on one another's evidence/reasoning.
- The final response is coherent.

Doesn't Look Like

- One student does all writing and talking.
- Contributions are disconnected facts.

- The group stops after a claim with no evidence.

Physics Hot Seat

Course Routines + CER

[← Activity Options](#)

LEARNING TARGET

I can successfully participate in physics class by following course procedures and using evidence-based reasoning to explain physical phenomena.

STRUCTURE

Team explanation → One speaker faces questions

SETUP

Groups prepare a response; one student becomes the first Hot Seat explainer.

DIRECTIONS

1. Team solves the projected problem together.
2. Hot Seat student explains the phenomenon, evidence, and reasoning without reading a script.
3. Class/team asks one clarifying or challenge question.
4. Rotate the Hot Seat for the next problem.

GOAL

Make the physics thinking visible: identify the phenomenon, use relevant evidence, and explain the physical relationship.

QUESTION SETS

[Set 1](#) · [Set 2](#)

Looks Like Success

- Speaker can explain why evidence matters.
- Questions target the physics, not the person.
- Hot Seat rotates.

Doesn't Look Like

- Speaker only gives the final answer.
- Questions become personal or distracting.
- Same student speaks every round.

Stations

Course Routines + CER

[← Activity Options](#)

LEARNING TARGET

I can successfully participate in physics class by following course procedures and using evidence-based reasoning to explain physical phenomena.

STRUCTURE

Teams rotate through six 4-question stations.

SETUP

Print the duplex station packet. Place one question page at each station; keep answer pages for checking/teacher use as desired.

DIRECTIONS

1. Every student attempts each problem.
2. Compare evidence/reasoning before changing an answer.
3. Rotate only when signaled.
4. Leave the station ready for the next group.

FILES

[Set 1 Stations](#) · [Set 2 Stations](#)

Looks Like Success

- All students are writing or discussing physics.
- Teams use data/phenomena when explaining.
- Rotations are orderly.

Doesn't Look Like

- One person completes the station for everyone.
- Groups copy answers without discussion.
- Materials are left disorganized.

Find Someone Who

Course Routines + CER

[← Activity Options](#)

LEARNING TARGET

I can successfully participate in physics class by following course procedures and using evidence-based reasoning to explain physical phenomena.

STRUCTURE

Individual attempt → Find a classmate → Compare and explain

SETUP

Each student receives one handout.

DIRECTIONS

1. Attempt a problem before moving.
2. Find a classmate and compare reasoning.
3. Each person explains one idea or correction.
4. Record initials only after both people can explain the response.
5. Move to a different classmate for the next problem.

FILES

[Set 1](#) · [Set 2](#)

Looks Like Success

- Students attempt before circulating.
- Partners discuss evidence/reasoning.
- Students meet several classmates.

Doesn't Look Like

- Walking around before attempting physics.
- Collecting initials without discussion.
- One student fills in another's sheet.